UNIVERSITY OF CAPE COAST
DEPARTMENT OF MATHEMATICS
MAT 101: ALGEBRA AND TRIGONOMETRY

TUTORIAL SET 5.

1. Convert the following from degrees to radians
   (a) 60°
   (b) -45°
   (c) 107°
   (d) 90°
2. Convert the following from radians to degree
   (a) $\frac{\pi}{6}$ radians.
   (b) $\frac{3\pi}{2}$ radians.
   (c) $\frac{-7\pi}{4}$ radians.
3. Given $sin\theta = \frac{\sqrt{5}}{5}$ and $cos\theta = \frac{2\sqrt{5}}{5}$, find the value of $cosec\theta$, $cot\theta$, $tan\theta$ and $secant\theta$.
4. Express $3cos\theta + 4sin\theta$ in the form $Rcos(\theta - \alpha)$ where R and α are constants, $R > 0$ and $0 < \alpha < 90^\circ$.
   (a) Hence find the maximum and minimum value of $3cos\theta + 4sin\theta$ and the corresponding values of θ.
   (b) Using your solution in above, solve $3cos\theta + 4sin\theta = \frac{5}{2}$. where $0 < \theta < 180^\circ$.

5(a). Use the double angle formulae and identity $cos(A + B) = cosAcosB - sinAsinB$ to obtain an expression for $cosx$ only.

(b) Prove that $\frac{cosx}{1+sinx} + \frac{1+sinx}{cosx} = 2secx$.

(c) Hence find, for $0 < x < 2\pi$, all solutions of $\frac{cosx}{1+sinx} + \frac{1+sinx}{cosx} = 4$.